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Glasma Role in Jet Quenching Effects

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Examination Committee

Chairperson: Prof. Full Name 1
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Month Year

Dedicated to someone special...

Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Acknowledgments

A few words about the university, financial support, research advisor, dissertation readers, faculty or other professors, lab mates, other friends and family...

Resumo

Inserir o resumo em Português aqui com um máximo de 250 palavras e acompanhado de 4 a 6 palavras-chave.

Palavras-chave: palavra-chave1, palavra-chave2, palavra-chave3,...

Abstract

Insert your abstract here with a maximum of 250 words, followed by 4 to 6 keywords.

Keywords: keyword1, keyword2, keyword3,...

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BFKL Balitsky–Fadin–Kuraev–Lipatov i, 4

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DIS Deep Inelastic Scattering i, 3

GRL-MQ Gribov–Riskin–Levin–Mueller–Qiu i, 5

HIC Heavy Ion Collision i, 5, 9, 15

IMF Infinite Momentum Frame i, 5, 6

JIMWLK Jalilian-Marian–Iancu–McLerran–Weigert–Leonidov–Kovner i, 9

MV McLerran-Venugopalan i, 9

PDF Parton Distribution Function i, xvii, 3, 4

QCD Quantum Chromodynamics i, 4–7

Chapter 1

Chapter 2

2.1 The QCD Lagrangian

In studying a system where only strong interactions are relevant, one turns to the QCD Lagrangian. It is a functional of the quark fields, ψ, which are 4-component Dirac spinors – with $\overline{\psi} = \psi^\dagger \gamma^0$ being the quark field's Dirac adjoint –, and of the gluon fields, A_μ^a, which are real-valued four-vectors; both are functions of spacetime. For a theory with N_c colors, considering a single flavor of quarks with mass m, the Lagrangian is

$$\mathcal{L}_{\text{QCD}} = \overline{\psi}\left(i\gamma^\mu D_\mu - m\right)\psi - \frac{1}{2}\operatorname{Tr}_c\left[F_{\mu\nu}F^{\mu\nu}\right] \tag{2.1}$$

where D_μ, the covariant derivative, is given by:

$$D_\mu = \partial_\mu - igA_\mu, \quad \text{where } A_\mu = A_\mu^a T_a \tag{2.2}$$

and $F^{\mu\nu}$, the gluon field strength tensor, is given by:

$$\begin{aligned} F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu] \equiv F_{\mu\nu}^a T_a \\ F_{\mu\nu}^a &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c \end{aligned} \tag{2.3}$$

where g is the Strong Force's coupling constant; T^a are the SU(N_c) generators in the fundamental representation[1], and f^{abc} are the SU(N_c) structure constants; Tr_c denotes a color-trace, that is, a trace in color space, the space of the T_a matrices' indices.

Writing the Lagrangian with all indices explicit,

$$\mathcal{L}_{\text{QCD}} = \left(\psi_\alpha^i\right)^* \left(\gamma^0\right)_{\alpha\beta} \left(i\left(\gamma^\mu\right)_{\beta\gamma}\left(\delta^{ij}\partial_\mu - igA_\mu^a\left(T^a\right)^{ij}\right) - \delta_{\beta\gamma}\delta^{ij}m\right)\psi_\gamma^j - \frac{1}{4}F_{\mu\nu}^a F_a^{\mu\nu} \tag{2.4}$$

where $\alpha, \beta, \gamma = 1, 2, 3, 4$ are Dirac indices, $i, j = 1, ..., N_c$ are color indices in the fundamental representation, $a = 1, ..., {N_c}^2 - 1$ are color indices in the adjoint representation, and $\mu, \nu = 0, 1, 2, 3$ are spacetime indices when using Minkowski coordinates.

To obtain the Lagrangian considering all flavors of quarks, one need only add copies of the first term of Eq. (2.1) for each quark flavor's quark fields.

[1] Normalized such that $\operatorname{Tr}_c[T^a, T^b] = \delta^{ab}/2$

Given the presented QCD Lagrangian, the corresponding Euler-Lagrange equations relating to the gluon fields A are called the Yang-Mills equations, which can be thought of as the QCD equivalent to Classical Electrodynamics' Maxwell's equations. In the absence of quark fields ($\psi = 0$), they read:

$$[D_\mu, F^{\mu\nu}] = 0 \tag{2.5}$$

2.2 [A picture of nuclear matter]

[Part of it may migrate. At some point before, introduce the proton as something more than 3 quarks]

Probing hadrons at high energies in scattering experiments has allowed exploring the interior structure of nucleons. Data from both fixed-target and collider (HERA, Tevatron, LHC) experiments has been compiled in order to obtain pictures of the parton (define) content of hadrons.

At HERA, electrons (or positrons) and protons were accelerated to collide at center of mass energies of around $\sqrt{s} = 300\,\mathrm{GeV}$. In some of these collisions, the momentum transfer Q^μ from the electron to the proton via a virtual photon was large enough ($Q^2 > 200\,\mathrm{GeV}^2$) to disintegrate the proton into a shower of hadrons $(e + p \to e + X)$; these are called Deep Inelastic Scattering (DIS) collisions, whose cross-sections are valuable in unveiling the proton's parton content. The greater the virtual photon's energy, the greater its ability to resolve structures of smaller scale than the proton radius and with shorter characteristic timescales. DIS collisions have also been studied for this purpose using muons, μ, and neutrinos, ν, as probes, and neutrons as targets. At the Tevatron and LHC colliders, data from $p\overline{p}$ and pp collisions, respectively, has contributed to the study of the gluon content of protons, specifically. [1][2]

The contributions from these many experiments have permitted the measurement of the proton's PDFs, which quantitatively characterize the parton content of a hadron. They translate the abundance, inside the proton, of a given parton species carrying a certain fraction of its momentum (in a frame of reference where the proton's momentum is large). The PDF of a species of parton q inside the proton, $q(x)$, is defined such that the number of q partons carrying a fraction of the proton's momentum between x and $x + \Delta x$ is given by $q(x)\Delta x$. These functions actually also depend on the energy scale of the process used to probe the proton, Q^2, since a greater exchange of energy allows probing finer features of the proton, increasing its PDFs at small x, for example. [1]

Fig. 2.1 shows the different parton species' PDFs for the proton. For $0.1 \lesssim x < 1$, the "valence" quarks u ($\mathbf{u}_V$)and d ($\mathbf{d}_V$) dominate – in the relative proportion expected from the static quark model for the proton, uud. However, for $x \lesssim 0.02$, they are overtaken by a "sea" of quarks and antiquarks ($\overline{\mathbf{u}}$, $\overline{\mathbf{d}}$, $\mathbf{s}$, $\mathbf{c}$, $\mathbf{b}$), which, for the same flavor, have identical PDFs. For small x, however, the most abundant parton is, by far, the gluon ($\mathbf{g}$).

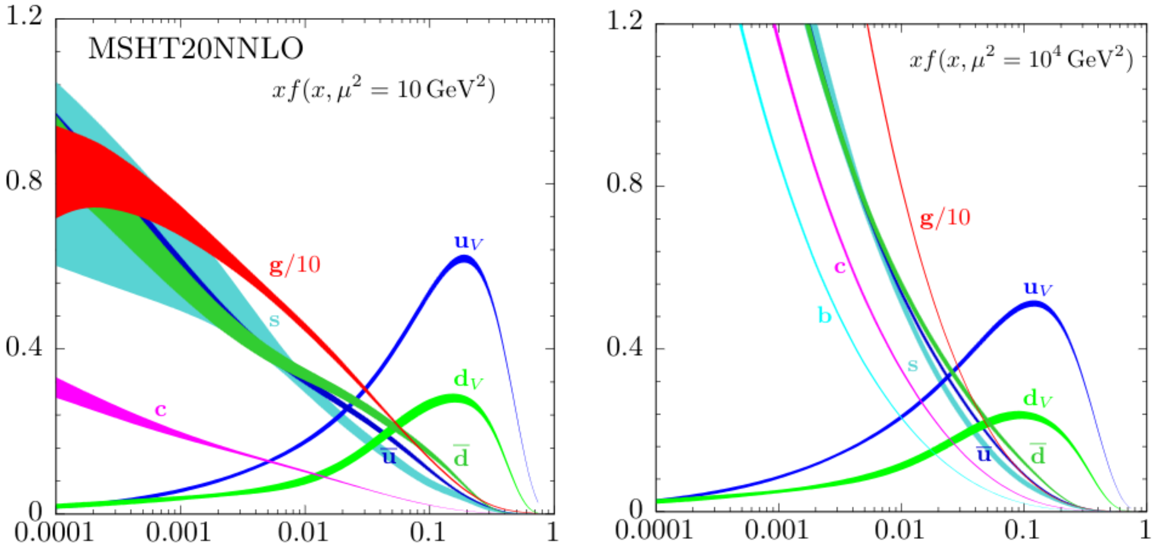

Figure 2.1: The proton's PDFs multiplied by x for the case of unpolarized beams at the energy scales $\mu^2 = 10\,\mathrm{GeV}^2$ (left) and $\mu^2 = 10^4\mathrm{GeV}^2$ (right). The gluon's PDF appears scaled down by a factor of 10. [2] Originally from and calculated by the MSHT Group. [3]

2.3 Gluon Saturation

This dominance of the gluon "sea" at low-x is predicted by Quantum Chromodynamics (QCD), as discussed here regarding the splitting function of the gluon and the Balitsky–Fadin–Kuraev–Lipatov (BFKL) equations.

The splitting function of non-Abelian gauge bosons such as the gluon is a well know QCD result. It returns the probability of emission by a parton of a gluon with relative transverse momentum $\mathbf{k}_\perp$ and a fraction x of its longitudinal momentum. At lowest order in α_S and for small x, it reads [4, 5]:

$$\mathrm{d}P \propto \frac{\alpha_S}{\pi^2}\frac{\mathrm{d}^2\mathbf{k}_\perp}{k_\perp}\frac{\mathrm{d}x}{x} \tag{2.6}$$

the singularity of this probability as $x \to 0$ suggests a picture of abundance of small-x gluons within all hadrons, which grows uncapped as partons emit small-x gluons, which themselves emit smaller-x gluons or produce quark-antiquark pairs which also serve as gluon emitters. However, gluon densities may not explode as $x \to 0$, as unitarity would be violated. [4]

Below a certain value of x, non-linear dynamics not considered so far should take over and tame the divergence, *saturating* the low-x gluon density in the hadron. The onset of saturation is controlled by the saturation scale, $Q_s(x)$, a transverse momentum scale below which non-linear effects become relevant. [4]

One interpretation of these non-linear effects is the onset, above certain gluon densities, of gluon recombination. Above that threshold, the probability of emission of a new gluon becomes comparable to the probability of recombination of gluons, stopping the growth of their density (Fig. 2.2).

More formally, the BFKL QCD evolution equation is a linear differential equation that governs how gluon density in momentum space within a hadron should evolve with x when that hadron is probed at a

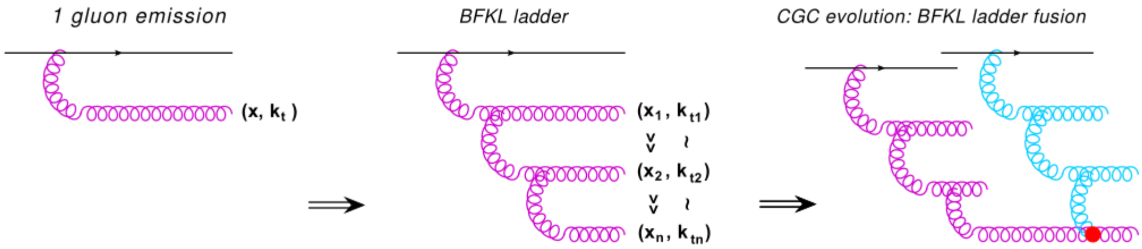

Figure 2.2: The onset of gluon recombination. Above a certain gluon density, gluon recombination (red dot) becomes comparatively likely to gluon emission (the pink and blue ladders of gluons). Taken from [4].

certain energy scale. It is a linear equation, and it predicts a divergence of the gluon density as $x \to 0$. The equation does not contain terms reflecting interference between emitters of gluons, as it assumes the hadron to be a dilute parton system. However, making the probability of emission of a new gluon dependent on the present gluon density allows for saturation to be reached when gluon generation and gluon recombination balance out. QCD evolution equations which attempt to incorporate this effect have additional, non-linear, density-dependent terms, such as the Gribov–Riskin–Levin–Mueller–Qiu (GRL-MQ) equation. [4]

2.4 The Color Glass Condensate and The McLarran-Venugopalan Model

Should I start with describing protons with the CGC and say it's the same for the nuclei? or should I start right away using the word nucleus?

The Color Glass Condensate (CGC) is an effective theory of QCD for high-energy QCD scattering applied to allow the description of the aforementioned saturated gluons [5], which evade standard perturbation techniques. It is one of the approaches used to study initial state effects in Heavy Ion Collisions (HICs) and has stood out due to its practical applicability and phenomenological success.[4] It hinges on the partition of the parton content of a hadron, in the Infinite Momentum Frame (IMF), via approximations, into static color charges and the dynamic classical gluon fields, representing the saturated gluons, which they generate via the processes described in Section 2.3.[5] In this case, the hadron in question is a heavy nucleus, which is made up of neutrons and protons whose contents are previously discussed.

2.4.1 Classical saturated gluon fields

YET UNJUSTIFIED: why $N \sim 1/g^2$; why g is small; do the creation operators create a gluon with a specific momentum $(x, \mathbf{k}_\perp)$?

Starting by justifying the treatment of the saturated gluons as classical fields, the saturated gluon densities at low-x discussed previously in Sections 2.2 and 2.3 imply gluon (or color) fields of the order

$A \sim 1/g$, which does not allow their treatment through standard perturbative techniques[2].[4] However, it opens the possibility of their treatment as classical fields.

Comparing the occupation number of the saturated gluon fields and the commutator of their creation, $a^\dagger$, and annihilation, a, operators, [4]

$$N = a^\dagger a \sim \frac{1}{g^2} \quad \gg \quad [a^\dagger, a] \sim 1 \tag{2.7}$$

it is possible to state that quantum corrections to the classical solution, where the fields commute, are negligible. Thus, it is reasonable to treat the low-x gluon fields as classical fields.

2.4.2 Partition of the nucleus's parton content

In order to illustrate the partition of the parton content of the nucleus into a static and a dynamic part, it is necessary to move to the infinite momentum frame. In the IMF, the nucleus is boosted along its direction of motion so as to make the momenta of its constituent partons approximately collinear to its own. In this frame, considering a nucleus moving in the positive x^1 direction with large total longitudinal momentum P much greater than its mass, m_A, the four-momenta of the nucleus and of one of its constituent partons in Minkowski coordinates are: [1]

$$\text{Nucleus: } P^\mu = (P, P, 0, 0) \qquad\qquad \text{Parton } i\text{: } p^\mu = (x_i P, x_i P, \sim 0, \sim 0) \tag{2.8a}$$

where $0 < x_i < 1$ is the fraction of the nuclear longitudinal momentum carried by the i-th parton; it was further assumed that the boost applied was such that $x_i P \gg m_i$, with m_i being the i-th parton's mass. In light-cone coordinates (Appendix B.1),

$$\text{Nucleus: } P^\mu = (P^+, 0, 0, 0) \qquad\qquad \text{Parton } i\text{: } p^\mu = (x_i P^+, \sim 0, \sim 0, \sim 0) \tag{2.8b}$$

where $P^+ = \sqrt{2}P$. It becomes clear that x_i is also the i-th parton's fraction of the nuclear light-cone $+$-momentum.

The uncertainty principle can be used to evaluate the approximate extent of each parton over the x^- and x^+ coordinates: [4, 6]

$$\Delta x^- \propto \frac{1}{p^+} = \frac{1}{x_i P^+} \tag{2.9a}$$

$$\Delta x^+ \propto \frac{1}{p^-} = \frac{2p^+}{(p^0)^2 - (p^1)^2} = \frac{2p^+}{(\mathbf{p}_\perp)^2 + {m_i}^2} \approx \frac{2x_i P^+}{(\mathbf{p}_\perp)^2} \tag{2.9b}$$

where $\mathbf{p}_\perp$ is the i-th parton's transverse momentum, which was taken to be negligible in Eqs. (2.8) when compared to its longitudinal momentum.

Two useful conclusions about the x^+, x^- scales of high-x and low-x partons may be extracted from the uncertainty relations:

[2]For example, in calculating the matrix element $\mathcal{M}$ for a QCD process using a series expansion in powers of g, all terms are proportional to $gA \sim \mathcal{O}(1)$, which does not allow truncating the series in order to estimate $\mathcal{M}$ – all orders need to be summed.

1) high-x partons are highly localized in the x^- direction. That is, they occupy a very thin "slice" of space in the x^- direction.

2) high-x partons are more delocalized in the x^+ direction than their low-x counterparts. That is, their typical x^+ scale is greater than the low-x partons'. Equivalently, the fluctuations of the low-x parton fields are much shorter along the x^+ direction than for the high-x parton fields.

? Include the naive [6] interpretations? 1) the Lorentz-contraction of the nucleus along x^1. 2) Time dilation experienced by valence partons.

It is from the second point that a hierarchy arises, central to the CGC formalism, of partons separated according to their x-values. A $+$-momentum scale, Λ^+, is introduced to separate the high-x, or valence, partons and the low-x partons. From the perspective of the latter, the first have very slow dynamics[3] in x^+; in the limit of this argument, they are *static* in x^+; for particles traveling at the speed of light, being static in x^+ is equivalent to being static in t. As such, in the CGC, valence partons become constant color charges, which serve as sources for the *dynamic* low-x parton fields. As was introduced in sections 2.2 and 2.3, the partonic content of protons, and hence of the nucleus, is dominated by gluons at low-x. Therefore, the dynamic fields are taken to be gluon, also called color, fields.

The parton content of the nucleus is thus divided into the group of high $+$-momentum ($p^+ > \Lambda^+$) static color charges and the group of low $+$-momentum ($p^+ < \Lambda^+$) dynamic classical gluon fields which they generate. The static color charges may themselves be treated as classical [4, 7] and are eikonally coupled to the classical gluon fields; that is, they suffer negligible recoil from their interactions with the color fields, when compared to their longitudinal momentum.[5] Thus, they are introduced into the QCD Lagrangian (Eq. (2.4)) as classical color current 4-vectors in the term $A^a_\mu J^\mu_a$. Since, now, the formalism does not include quantum quark fields, Eqs. 2.1 and 2.4 become, respectively,

$$\mathcal{L}'_{\text{QCD}} = -\frac{1}{2}\,\mathrm{Tr}_c\left[F_{\mu\nu}F^{\mu\nu}\right] + 2\,\mathrm{Tr}_c[A_\mu J^\mu] \tag{2.10}$$

$$\mathcal{L}'_{\text{QCD}} = -\frac{1}{4}F^a_{\mu\nu}F^{\mu\nu}_a + A^a_\mu J^\mu_a \tag{2.11}$$

where $J^\mu \equiv J^\mu_a T^a$.

Consequently, the corresponding Euler-Lagrange equations for the gluon fields A become:

$$[D_\mu, F^{\mu\nu}] = J^\nu \tag{2.12}$$

which, provided an expression for the external color currents, may be solved for the classical gluon fields.

Seeing J^μ as a classical color current describing the high-x partons, which travel at near the speed of light in the x^1 direction, then, in light-cone coordinates, its only non-vanishing component is J^+ (Eq. (2.13)). Additionally, it will directly depend on the number color charge distribution of the valence partons,

[3] Also slow in the timescale of the pre-hydrodynamic stage?

$\rho(x) \equiv \rho_a(x)T^a$ (Eq. (2.14)).

$$J^\mu = \delta^\mu_+ J^+ \tag{2.13}$$

$$J^+ = g\rho(x^\mu) \tag{2.14}$$

From the previous discussion on the static nature of the color charges, ρ's dependence on x^+ can be dropped. Furthermore, due to the first conclusion drawn from the uncertainty relations (Eqs. 2.9), the color charge distribution function describing the placement of the static valence partons is very thin in the x^- direction. In the limit where this thin x^- dependence factorizes to a Dirac-delta function, the valence color current four-vector is, in light-cone coordinates:

$$J^\mu = (g\delta(x^-)\rho(\mathbf{x}_\perp), \sim 0, \sim 0, \sim 0) \tag{2.15}$$

where the charge distribution has been taken to be longitudinally homogenous and ρ has become an area number charge density.

2.4.3 The stochastic nature of ρ

The color charge distribution ρ, from which J^μ and, consequently, the evolution of the gluon fields are obtained, is fixed for a single nucleus in a single event. It is different, however, for different nuclei and for each event, constituting a stochastic variable [5] which isn't experimentally accessible. Even for the same nucleus, ρ would change if enough time is allowed to pass in order for the nucleus to travel over a x^+ greater than the valence partons' typical x^+ scale [8]. In order to calculate an observable $\mathcal{O}$ out of many events within the CGC effective theory, one must determine its average value considering all the different possible color charge distributions and their probabilities: [5]

$$\langle \mathcal{O} \rangle = \int [\mathrm{d}\rho] W[\rho] \mathcal{O}[\rho] \tag{2.16}$$

where $\mathcal{O}[\rho]$ is the observable's value for the particular charge distribution ρ, and $W[\rho]$ is a functional probability distribution evaluated for that same charge distribution, describing its probability of occurring.

2.4.4 Quantum Corrections and the McLerran-Venugopalan Model

So far, the effect of the choice of a $+$-momentum scale Λ^+ to separate low- and high-x partons on the final result of an observable, $\langle \mathcal{O} \rangle$, has been neglected. Indeed, once a $+$-momentum scale has been set, one neglects that certain low-x partons sorted into the category of classical gluon fields actually behave as static color charges for lower-x-still partons[4] [4]. Thus, choosing a different Λ^+ would lead to different observable values. When changing the value of the $+$-momentum scale from Λ^+ to $\Lambda^{+\prime}$, it is possible, however, to renormalize ρ, correlators of different ρ or the functional probability distribution W in order to incorporate those quantum corrections into the CGC effective theory and preserve the value

[4] In the language of Section 2.3, even a low-x gluon can serve as an emitter of lower-x gluons.

of the observables, making them independent of Λ^+, while keeping the calculation procedure strictly classical [4]. Choosing to incorporate these quantum corrections into the probability distribution, which then becomes dependent on the $+$-momentum scale, W_{Λ^+}, its evolution with Λ^+ is ruled by a functional Jalilian-Marian–Iancu–McLerran–Weigert–Leonidov–Kovner (JIMWLK) equation [5]. Other methods to incorporate quantum corrections into the effective theory exist, as well as variations on the JIMWLK equation [4].

In spite of this achievement, an initial probability function $W_{\Lambda_0^+}$ is still necessary before studying its evolution with the $+$-momentum scale; this function can be seen as an initial condition for the JIMWLK equation. The McLerran-Venugopalan (MV) model provides that initial condition, for the case of large nuclei. It is physically motivated and may even be used directly, without needing to be evolved into x-scales much smaller than Λ_0^+, for the study of pA or AA collisions where the values of x are not small enough to require that evolution. [5] W_{MV}, not normalized, is given by [8]:

$$W[\rho_a] = \exp\left[-\int \mathrm{d}^2\mathbf{x}_\perp \frac{\rho_a(\mathbf{x}_\perp)\rho_a(\mathbf{x}_\perp)}{2\mu^2}\right] \tag{2.17}$$

which has an associated charge density 2-point correlator:

$$\langle \rho_a(\mathbf{x}_\perp)\rho_b(\mathbf{y}_\perp)\rangle = \mu^2\delta_{ab}\delta^2(\mathbf{x}_\perp - \mathbf{y}_\perp) \tag{2.18}$$

How do I relate these two?

[Some authors use $\mu(x^-)$, $\rho(x^-, \mathbf{x}_\perp)$ for W [4] or for the 2-point correlator [6]. From what I saw, only [8] introduces the MV model without the x^- coordinate here, but for $W[\rho_a]$ instead of $W[\rho]$.][Should I not factorize the x^- as soon as I do, so that I can use those results here?]

The MV model's functional probability distribution W_{MV} and corresponding charge density 2-point correlator are motivated physically for the case of large nuclei: [8]

1) Due to color confinement, color charge within different nucleons should be uncorrelated. Since charge densities at different transverse positions are associated with different nucleons/longitudinal superpositions of nucleons, they should also be uncorrelated.

2) For large nuclei, at least at central positions, transverse color charge density results from the sum of many longitudinally aligned nucleons. Since the color charge of those nucleons is uncorrelated, then $\rho_a(\mathbf{x}_\perp)$ is a sum of uncorrelated variables. Therefore, the Central Limit Theorem applies, and $\rho_a(\mathbf{x}_\perp)$ should follow a Gaussian distribution.

In this work, the non-evolved MV model will be used. The application of this formalism to the two-nuclei system of a HIC will be introduced over the next sections. It was first accomplished in [9].

2.5 [Setting the stage/introducing the initial situation]

In a HIC, as previously introduced, two heavy nuclei travel in opposite directions at near the speed of light before crashing into each other. For the purposes of this work, both nuclei will be taken to travel at the speed of light; because of this, they suffer length contraction along the direction of their trajectories, becoming infinitely thin/planar[5]. Furthermore, their collision is taken to be central, that is, the trajectories of their centers are perfectly aligned.

In the laboratory/center of mass frame, the x^1 axis is placed along the trajectories of the nuclei. The nucleus moving in the positive x^1 direction is identified as nucleus 1, and the one moving in the negative x^1 direction is identified as nucleus 2. The nuclei collide at $x^1 = 0$, at instant $t = 0$. This system is invariant under boosts along the x^1 direction.

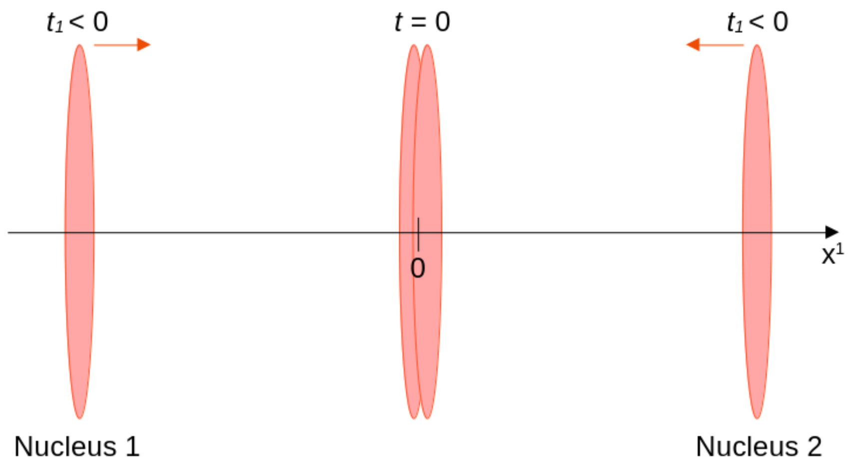

Figure 2.3: Two heavy nuclei, Lorentz-contracted into thin disks, travel along the x^1 axis in opposite directions until they collide at $t = 0, x^1 = 0$.

As such, prior to the collision, each nucleus's position four-vector, $x^\mu = (x^0, x^1, x^2, x^3)$, in Minkowski coordinates, is:

$$\text{Nucleus 1}: \quad (t, t, 0, 0)$$
$$\text{Nucleus 2}: \quad (t, -t, 0, 0)$$

These trajectories are well-illustrated in the $x^1 O x^0$ plane (Fig. 2.4).

After the collision, the nuclei's approximately static high-x partons, discussed in Section 2.4.2, carry on in the direction they were moving at the speed of light, acting as color sources for the gluon fields then generated in the space between the now receding nuclei. It is in this space that the pre-hydrodynamic stage to be studied is formed. Fig. 2.4 already suggests the usefulness of the ligh-cone coordinate system (see Appendix B.1), as the static color sources only travel along the edges of the light-cone centered on the collision coordinates $(0, 0)$. In this coordinate system, the nuclei's position four-vectors, $x^\mu = (x^+, x^-, x^2, x^3)$, are:

$$\text{Nucleus 1}: \quad (\sqrt{2}t, 0, 0, 0)$$
$$\text{Nucleus 2}: \quad (0, \sqrt{2}t, 0, 0)$$

and the collision coordinates remain at $(x^+ = 0, x^- = 0)$.

[5] One reference makes a comparison between the longitudinal extent of the nucleus and the energy of the probe??

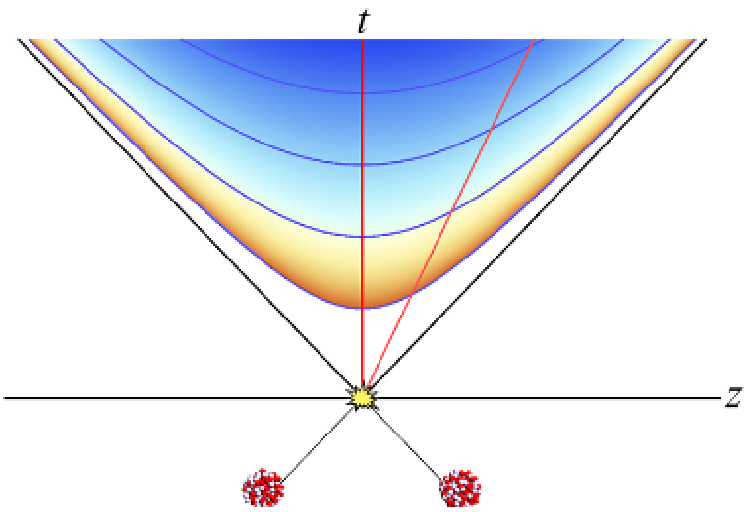

Figure 2.4: A heavy-ion collision represented in the x^1Ox^0 plane. Both nuclei travel along the light-cone centered around the collision at coordinates $(0,0)$. Taken from [10].

In order to fulfill this work's objectives, it will be necessary to calculate the gluon fields generated in the space between the receding nuclei, making use of the classical Yang-Mills equations. The fields will be determined from the the static distributions of color charge of the receding nuclei.

Next: event averages? Move the introduction to dEdx and $\hat{q}$ to before this? say they are the result of the color fields acting on the quark probe?

2.6 Solving The Yang-Mills Equations

2.6.1 One nucleus

It is easiest to start by considering the case of a single nucleus, chosen to be Nucleus 1, moving in the positive x^1 direction. Solutions to the Yang-Mills equations for a single nucleus moving along the light-cone were first found analytically in [11] for the classical case $D_\mu F^{\mu\nu} = J^\nu$. Then maybe cite where it was made in the light-cone gauge from the beginning.

The Yang-Mills equations are initially solved in the Lorenz, or covariant, gauge, where $\partial_\mu A^\mu = 0$, following the more recent solving of the full equations from [12, Section 2.2]. The solution is then brought to light-cone gauge, where $A^+ = 0$, an axial gauge condition, following the procedure from [13, Section 3A]. ([9]: "one has direct access to the gluon distribution functions of the parton model"). I was going to write something here. Maybe the residual gauge freedom?

First of all, the equation explicitly enforcing the covariant conservation of color charge is added to the list of equations to solve. It is analogous to Classical Electromagnetism's equation of conservation of electric charge, and is obtained from the Yang-Mills equations as:

$$[D_\mu, F^{\mu\nu}] = J^\nu \implies [D_\nu, J^\nu] = [D_\nu, [D_\mu, F^{\mu\nu}]] = 0 \tag{2.19}$$

via the Jacobi identity and the $F^{\mu\nu} = -ig[D_\mu, D_\nu]$ identity for the gluon field strength tensor (see Appendix A.1 for a detailed derivation).

The conservation equation and The Yang-Mills equations (in the Lorenz gauge) are then arranged into the system of equations:

$$\begin{cases} [D_\mu, J^\mu] = 0 \\ [D_\mu, F^{\mu\nu}] = J^\nu \end{cases} \Leftrightarrow \begin{cases} \partial_\mu J^\mu = ig[A_\mu, J^\mu] \\ \partial_\mu \partial^\mu A^\nu = J^\nu + ig[A_\mu, F^{\mu\nu} + \partial^\mu A^\nu] \end{cases} \tag{2.20}$$

A solution may be built by solving this system iteratively, for the fields and current expanded in terms of powers of the color number charge density, ρ. The terms of order (n) are obtained from those of lower orders by:

$$\begin{cases} \partial_\mu J^\mu_{(n)} = ig[A_\mu, J^\mu]_{(n)} \\ \partial_\mu \partial^\mu A^\nu_{(n)} = J^\nu_{(n)} + ig[A_\mu, F^{\mu\nu} + \partial^\mu A^\nu]_{(n)} \end{cases} \tag{2.21}$$

Inputting $J^\mu_{(0)} = A^\mu_{(0)} = 0$ (since, in the absence of color charges, there is neither a field generated from them nor any color currents) and $J^\mu_{(1)} = g\delta^\mu_+ \rho(x^-, \mathbf{x}_\perp)$ (Eq. (2.15)'s form for J^μ), the top equation for $n = 1$ is automatically satisfied since $\partial_+ J^+_{(1)} = 0$. The bottom equation may be solved by requiring that $A^\mu_{(1)}$, like $J^\mu_{(1)}$, be independent of x^+, and that it vanishes in the far past/does not explode in the far past?:

$$\begin{aligned} \partial_\mu \partial^\mu A^\nu_{(1)} = J^\nu_{(1)} &\rightarrow -\nabla^2_\perp A^\nu_{(1)} = J^\nu_{(1)} \\ &\rightarrow A^\nu_{(1)}(x^-, \mathbf{x}_\perp) = -g\delta^\nu_+ \int \mathrm{d}^2\mathbf{y}_\perp G(\mathbf{x}_\perp - \mathbf{y}_\perp)\rho(x^-, \mathbf{y}_\perp) \end{aligned} \tag{2.22}$$

where $G(\mathbf{x}_\perp - \mathbf{y}_\perp)$ is a Green's function for the 2-dimensional Laplacian, $\nabla^2_\perp = \partial_i \partial^i$ with $i = 1, 2$.

Moving to the $n = 2$ system, the right side of the top equation is once again 0 since $A^\mu_{(1)}$ and $J^\mu_{(1)}$ only have a $+$-component. For the bottom equation, its commutator vanishes because $A^\mu_{(1)}$ only has a $+$-component and because that component is independent of x^+. The system for $n = 2$ is, then,

$$\begin{cases} \partial_\mu J^\mu_{(2)} = 0 \\ \partial_\mu \partial^\mu A^\nu_{(2)} = J^\nu_{(2)} \end{cases} \tag{2.23}$$

Once again requiring that the current and fields vanish in the far past/do not explode in the far past?, the solution is $J^\mu_{(2)} = 0$ for the top and $A^\mu_{(2)} = 0$ for the bottom. This makes all the greater-order terms also 0, so the final solution becomes:

$$J^\mu = g\delta^\mu_+ \rho(x^-, \mathbf{x}_\perp) \tag{2.24a}$$

$$A^\nu(x^-, \mathbf{x}_\perp) = -g\delta^\nu_+ \int \mathrm{d}^2\mathbf{y}_\perp G(\mathbf{x}_\perp - \mathbf{y}_\perp)\rho(x^-, \mathbf{y}_\perp) \tag{2.24b}$$

Finally, in order to move to light-cone gauge, where $A^+_{\text{LC}} = 0$, the appropriate gauge transformation matrix U must be found:

$$A^{\mu}_{\text{LC}} = U^{\dagger}A^{\mu}U + \frac{i}{g}U^{\dagger}\partial^{\mu}U \tag{2.25}$$

It may be found by solving the equation:

$$A^{+}_{\text{LC}} = 0 \;\Leftrightarrow\; U^{\dagger}A^{+}U + \frac{i}{g}U^{\dagger}\partial^{+}U = 0 \;\Leftrightarrow\; A^{+}U + \frac{i}{g}\partial_{-}U = 0 \;\Leftrightarrow\; \partial_{-}U = igA^{+}U \tag{2.26}$$

which admits a solution of the type $U(x^{-}, \mathbf{x}_{\perp})$ and may be solved iteratively in its integral form:

$$U(x^{-}, \mathbf{x}_{\perp}) = U(x_0^{-}, \mathbf{x}_{\perp}) + ig\int_{x_0^{-}}^{x^{-}} \mathrm{d}z A^{+}(z, \mathbf{x}_{\perp})U(z, \mathbf{x}_{\perp}) \tag{2.27}$$

$$\begin{aligned}
U(x^{-}, \mathbf{x}_{\perp}) &= \lim_{n\to\infty}\Bigg(\mathbb{1} + ig\int_{x_{-}^{0}}^{x^{-}} \mathrm{d}z_0 A^{+}(z_0) + (ig)^2\int_{x_{-}^{0}}^{x^{-}} \mathrm{d}z_0 \int_{x_{-}^{0}}^{z_0} \mathrm{d}z_1 A^{+}(z_0)A^{+}(z_1) + \ldots \\
&\qquad + (ig)^n \int_{x_{-}^{0}}^{x^{-}} \mathrm{d}z_0 \int_{x_{-}^{0}}^{z_0} \mathrm{d}z_1 \ldots \int_{x_{-}^{0}}^{z_{n-1}} \mathrm{d}z_n\, A^{+}(z_0)A^{+}(z_1)\ldots A^{+}(z_n)\Bigg)U(x_0^{-}, \mathbf{x}_{\perp}) \\
&= \lim_{n\to\infty}\Bigg(\mathbb{1} + ig\int_{x_{-}^{0}}^{x^{-}} \mathrm{d}z_0 A^{+}(z_0) + \frac{(ig)^2}{2}\int_{x_{-}^{0}}^{x^{-}} \mathrm{d}z_0 \int_{x_{-}^{0}}^{x^{-}} \mathrm{d}z_1 \mathcal{P}^{-}\Big\{A^{+}(z_0)A^{+}(z_1)\Big\} + \ldots \\
&\qquad + \frac{(ig)^n}{n!} \int_{x_{-}^{0}}^{x^{-}} \mathrm{d}z_0 \int_{x_{-}^{0}}^{x^{-}} \mathrm{d}z_1 \ldots \int_{x_{-}^{0}}^{x^{-}} \mathrm{d}z_n\, \mathcal{P}^{-}\Big\{A^{+}(z_0)A^{+}(z_1)\ldots A^{+}(z_n)\Big\}\Bigg)U(x_0^{-}, \mathbf{x}_{\perp}) \\
&\equiv \mathcal{P}^{-}\exp\left[ig\int_{x_0^{-}}^{x^{-}} \mathrm{d}z A^{+}(z, \mathbf{x}_{\perp})\right]U(x_0^{-}, \mathbf{x}_{\perp})
\end{aligned} \tag{2.28}$$

where the dependence of the A^{+} fields on $\mathbf{x}_{\perp}$ was only made explicit in the last line and $\mathcal{P}^{-}$ is the path-ordering operator for the x^{-} variable; that is, it orders the A^{+} field operators from right to left in descending order of their x^{-} argument:

$$x_n^{-} \geq x_{n-1}^{-} \geq \ldots \geq x_1 \implies \mathcal{P}^{-}\left\{A^{+}(x_{p_n}^{-})A^{+}(x_{p_{n-1}}^{-})\ldots A^{+}(x_{p_1})\right\} \equiv A^{+}(x_n^{-})A^{+}(x_{n-1}^{-})\ldots A^{+}(x_1) \tag{2.29}$$

with $\{p_n, p_{n-1}, \ldots, p_1\}$ some permutation of $\{n, n-1, \ldots, 1\}$

Additionally, $\mathcal{P}^{-}\exp[ig\int_{x_0^{-}}^{x^{-}} \mathrm{d}z A^{+}(z, \mathbf{x}_{\perp})]$ is called the path-ordered exponential of the $ig\int_{x_0^{-}}^{x^{-}} \mathrm{d}z A^{+}(z, \mathbf{x}_{\perp})$ integral.

It is possible to choose x_0^{-} to be $-\infty$ and $U(x_0^{-}, \mathbf{x}_{\perp})$ to be $\mathbb{1}$ with U remaining a solution (Is this a part of residual gauge freedom?:

$$U(x^{-}, \mathbf{x}_{\perp}) = \mathcal{P}^{-}\exp\left[ig\int_{-\infty}^{x^{-}} \mathrm{d}z A^{+}(z, \mathbf{x}_{\perp})\right] \equiv W(x^{-}, \mathbf{x}_{\perp}) \tag{2.30}$$

which corresponds to a Wilson Line, $W(x^{-}, \mathbf{x}_{\perp})$.

Renaming the $+$-component of A in the Lorenz gauge to $\Phi(x^{-}, \mathbf{x}_{\perp})$, the expressions for the gluon fields in light-cone gauge become:

$$A^+_{\text{LC}} = 0 \tag{2.31a}$$

$$A^-_{\text{LC}} = 0 \tag{2.31b}$$

$$A^i_{\text{LC}} = \frac{i}{g} U(x^-, \mathbf{x}_\perp)^\dagger \, \partial^i U(x^-, \mathbf{x}_\perp), \qquad i = 1, 2 \tag{2.31c}$$

$$\text{with } U(x^-, \mathbf{x}_\perp) \;= \mathcal{P}^- \exp\left[ig \int_{-\infty}^{x^-} \mathrm{d}z \; \Phi(z, \mathbf{x}_\perp) \right]$$
$$= \mathcal{P}^- \exp\left[ig \int_{-\infty}^{x^-} \mathrm{d}z \; (-g) \int \mathrm{d}^2\mathbf{y}_\perp G(\mathbf{x}_\perp - \mathbf{y}_\perp) \rho(z, \mathbf{y}_\perp) \right]$$

where the A^i fields are called pure-gauge fields due to them arising solely out of the gauge transformation term involving only U (Eq. (2.26)).

One last topic that should be discussed before moving on to the case of two nuclei is that, unlike in the classical abelian case of Electrodynamics, in the non-abelian gauge theory of Classical Chromodynamics gauge transformations affect the distributions of color charge; otherwise, the introduced Lagrangian, Eq. 2.11, would not be invariant under SU(3) gauge transformations. Thus, one should remember that the color charge distribution ρ discussed in this chapter and present in its final result (Eqs. 2.31) is the color charge distribution in the Lorenz, and not the light-cone, gauge. In the rest of this work, the subscript "LC" denoting results in the light-cone gauge will be dropped; in order to avoid confusion, color number charge densities in the Lorenz gauge will be denoted by $\tilde{\rho}$ instead of ρ.

TALK ABOUT THE APPROACH STARTING IN THE LIGHT-CONE GAUGE?

2.6.2 Two nuclei

The previous discussion of the fields generated by a single nucleus traveling at the speed of light in the positive x^1 direction (Nucleus 1) can be easily adapted for a single nucleus traveling in the negative x^1 direction (Nucleus 2). Labeling all quantities in this scenario with the subscript 2, equivalent arguments to those used in Section 2.4.2 lead to a color current with only a $-$-component, of the form:

$$J^\mu_2 = g\delta^\mu_- \tilde{\rho}_2(x^+, \mathbf{x}_\perp) \approx g\delta^\mu_- \delta(x^+) \tilde{\rho}_2(\mathbf{x}_\perp) \tag{2.32}$$

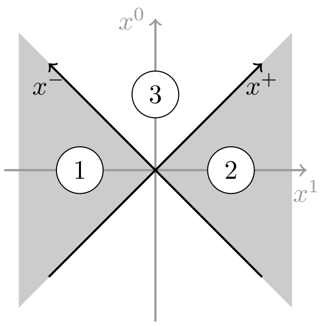

Figure 2.5: Partition of spacetime into light-cone quadrants. The quadrants where the Yang-Mills equations will be solved are identified as quadrants 1, 2 and 3. Adapted from [9].

Which generates the gluon fields, in the light-cone gauge:

$$A_2^+ = 0 \tag{2.33a}$$

$$A_2^- = 0 \tag{2.33b}$$

$$A_2^i = \frac{i}{g} U_2(x^+, \mathbf{x}_\perp)^\dagger\, \partial^i U_2(x^+, \mathbf{x}_\perp), \qquad i = 1, 2 \tag{2.33c}$$

$$\text{with } U_2(x^+, \mathbf{x}_\perp) = \mathcal{P}^+ \exp\left[ig \int_{-\infty}^{x^+} \mathrm{d}z\ \Phi_2(z, \mathbf{x}_\perp)\right]$$
$$= \mathcal{P}^+ \exp\left[ig \int_{-\infty}^{x^+} \mathrm{d}z\ (-g) \int \mathrm{d}^2\mathbf{y}_\perp G(\mathbf{x}_\perp - \mathbf{y}_\perp)\tilde{\rho}_2(z, \mathbf{y}_\perp)\right]$$

Moving to the scenario where both nuclei are present – the HIC –

In light cone coordinates, for nucleus 1, the only non-vanishing component of the color current four-vector J^ν is the $+$ component, ${J_1}^+$ (see Eq. (2.15)); for nucleus 2, by equivalent arguments applied to a nucleus moving in the opposite direction, the only non-vanishing component is the $-$ component, ${J_2}^-$, which has a similar form to ${J_1}^+$, but exchanging $x^- \to x^+$ [6]: gauge-dependent charge density

$$ {J_1}^+ = g\delta(x^-)\rho_1(\mathbf{x}_\perp) \qquad {J_2}^- = g\delta(x^+)\rho_2(\mathbf{x}_\perp)$$

where $\rho_{1,2}$ are the static number color densities in nuclei 1 and 2 per unit of transverse area.

Adding the contributions of both color currents, the source term becomes:

$$J^\nu = g\delta^\nu_+\delta(x^-)\rho_1(\mathbf{x}_\perp) + g\delta^\nu_-\delta(x^+)\rho_2(\mathbf{x}_\perp) \tag{2.34}$$

The classical Yang-Mills equations to be solved in the presence of the static color sources become, then,

$$[D_\mu, F^{\mu\nu}] = J^\nu \tag{2.35}$$

[6][14] [8] allows the nuclei to retain some thickness and then brings it to 0 at the end of the calculation. It justifies the use of ρ as a surface charge density, I think.

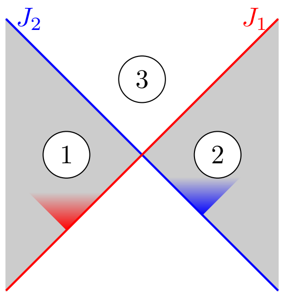

Figure 2.6

Attention: [9] initially posits that A^+ and A^- are different

$$
\begin{aligned}
A^+(x^\mu) &= \underbrace{\theta(x^+)\theta(x^-)}_{③}\, x^+ \alpha(\tau, \mathbf{x}_\perp) \\
A^-(x^\mu) &= -\underbrace{\theta(x^+)\theta(x^-)}_{③}\, x^- \alpha(\tau, \mathbf{x}_\perp) \\
A^i(x^\mu) &= \underbrace{\theta(-x^+)\theta(x^-)}_{①}\, \alpha_1^i(\mathbf{x}_\perp) + \underbrace{\theta(x^+)\theta(-x^-)}_{②}\, \alpha_2^i(\mathbf{x}_\perp) + \underbrace{\theta(x^+)\theta(x^-)}_{③}\, \alpha^i(\tau, \mathbf{x}_\perp)
\end{aligned}
\tag{2.36}
$$

2.7 Transcribed expressions yet without text

$$
\begin{aligned}
E_0^\eta(\mathbf{x}_\perp) &= -ig\delta^{ij}[\alpha_1^i(\mathbf{x}_\perp), \alpha_2^j(\mathbf{x}_\perp)] \\
&= g\delta^{ij}\alpha_{1a}^i(\mathbf{x}_\perp)\alpha_{2b}^j(\mathbf{x}_\perp) f^{abc} T_c
\end{aligned}
\tag{2.37a}
$$

$$
\begin{aligned}
B_0^\eta(\mathbf{x}_\perp) &= -ig\varepsilon^{ij}[\alpha_1^i(\mathbf{x}_\perp), \alpha_2^j(\mathbf{x}_\perp)] \\
&= g\varepsilon^{ij}\alpha_{1a}^i(\mathbf{x}_\perp)\alpha_{2b}^j(\mathbf{x}_\perp) f^{abc} T_c
\end{aligned}
\tag{2.37b}
$$

$$E^\eta(\tau, \mathbf{k}_\perp) = E_0^\eta(\mathbf{k}_\perp) J_0(k\tau) \tag{2.38a}$$

$$E^i(\tau, \mathbf{k}_\perp) = -i\varepsilon^{ij}\hat{k}^j B_0^\eta(\mathbf{k}_\perp) J_1(k\tau) \tag{2.38b}$$

$$B^\eta(\tau, \mathbf{k}_\perp) = B_0^\eta(\mathbf{k}_\perp) J_0(k\tau) \tag{2.38c}$$

$$B^i(\tau, \mathbf{k}_\perp) = -i\varepsilon^{ij}\hat{k}^j E_0^\eta(\mathbf{k}_\perp) J_1(k\tau) \tag{2.38d}$$

In order to get expressions for one of the fields in coordinate space[7], the appropriate field at $\tau = 0^+$ (eqs. 2.37) must be Fourier-transformed and used in Eqs. (2.38). The inverse Fourier transform is then applied to the resulting expression. For E^α, $\alpha = 1, 2$ (2,3)?, for example:

[7]as is necessary in order to calculate $X^{\alpha\beta}$

$$
\begin{aligned}
E^\alpha(\tau, \mathbf{x}_\perp) = &\int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)} e^{i\mathbf{k}_\perp\cdot\mathbf{x}_\perp}\,(-i)\varepsilon^{\alpha\beta}\hat{k}^\beta J_1(k\tau)\int \mathrm{d}^2\mathbf{u}_\perp e^{-i\mathbf{k}_\perp\cdot\mathbf{u}_\perp} \\
&\qquad\qquad\qquad\qquad \times\ g\varepsilon^{ij}\alpha^i_{1a}(\mathbf{u}_\perp)\alpha^j_{2b}(\mathbf{u}_\perp)f^{abc}T_c \\
= &-ig\varepsilon^{ij}f^{abc}T_c\varepsilon^{\alpha\beta}\int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)}\hat{k}^\beta J_1(k\tau)\int \mathrm{d}^2\mathbf{u}_\perp e^{i\mathbf{k}_\perp\cdot(\mathbf{x}-\mathbf{u})_\perp} \\
&\qquad\qquad\qquad\qquad \times\alpha^i_{1a}(\mathbf{u}_\perp)\alpha^j_{2b}(\mathbf{u}_\perp)
\end{aligned}
\tag{2.39}
$$

And the remaining fields:

$$
E^\eta(\tau, \mathbf{x}_\perp) = g\delta^{ij}f^{abc}T_c\int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)}J_0(k\tau)\int \mathrm{d}^2\mathbf{u}_\perp e^{i\mathbf{k}_\perp\cdot(\mathbf{x}-\mathbf{u})_\perp}\alpha^i_{1a}(\mathbf{u}_\perp)\alpha^j_{2b}(\mathbf{u}_\perp) \tag{2.40}
$$

$$
B^\eta(\tau, \mathbf{x}_\perp) = g\varepsilon^{ij}f^{abc}T_c\int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)}J_0(k\tau)\int \mathrm{d}^2\mathbf{u}_\perp e^{i\mathbf{k}_\perp\cdot(\mathbf{x}-\mathbf{u})_\perp}\alpha^i_{1a}(\mathbf{u}_\perp)\alpha^j_{2b}(\mathbf{u}_\perp) \tag{2.41}
$$

$$
B^\alpha(\tau, \mathbf{x}_\perp) = -ig\delta^{ij}f^{abc}T_c\varepsilon^{\alpha\beta}\int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)}\hat{k}^\beta J_1(k\tau)\int \mathrm{d}^2\mathbf{u}_\perp e^{i\mathbf{k}_\perp\cdot(\mathbf{x}-\mathbf{u})_\perp}\alpha^i_{1a}(\mathbf{u}_\perp)\alpha^j_{2b}(\mathbf{u}_\perp) \tag{2.42}
$$

Chapter 3

[Margaret]

[Likely to Migrate, requires references (textbooks?)]

The objectives of this work are: the derivation, by analytical means, of expressions for parameters characterizing the pre-equilibrium stage of a HIC as a medium which affects the highly energetic partons that traverse it; and the numerical evaluation of those expressions and comparison of the results obtained to others in the literature.

Those parameters are: the *stopping power*, denoted by $-\,\mathrm{d}dE/\,\mathrm{d}x$, and the *momentum broadening coefficient*, denoted by $\hat{q}$.

In Particle Physics, the stopping power of a medium as felt by a probe traversing it at a given instant corresponds to the amount of kinetic energy being lost by the probe per unit length traveled in the medium. Less formally, it is equal to the infinitesimal loss of energy of the probe by interaction with the medium divided by the infinitesimal distance traversed during that loss:

$$-\frac{\mathrm{d}E}{\mathrm{d}x} \approx -\frac{\Delta E}{\Delta x}$$

The stopping power may be viewed as a measure of how capable a medium is of slowing down or stopping traversing particles within a given length, which makes it a quantity commonly discussed in the field of radiation protection, for example.

The momentum broadening coefficient, on the other hand, measures how capable a medium is of deflecting traversing particles, that is, of changing their original direction. It corresponds to the square magnitude of transverse momentum gained by the probe per unit length traveled in the medium, transverse meaning perpendicular to the probe's velocity in that instant.

$$\hat{q} \approx \frac{(\Delta \mathbf{p}_T)^2}{\Delta x}$$

These quantities are dependent on the properties of the medium and of the probe, on the velocity of the probe, and, in the case of a heterogeneous, dynamic medium, on the current position of the probe and on time.

The chromodynamic field correlators derived in the previous section may be used to obtain the momentum broadening coefficient, $\hat{q}$, and the stopping power, $\mathrm{d}E/\mathrm{d}x$, felt by a relativistic quark probe traversing the glasma. This procedure is outlined in [13].

3.1 The Fokker-Planck equation and its collision terms

The Fokker-Planck equation has frequently been used to study the transport of heavy quarks across a thermalized QGP. The following Fokker-Planck equation, derived in [15] for the transport of heavy quarks through a glasma, is also applicable to relativistic light partons, as long as the diffusion approximation is applicable [13], that is, as long as the transfer of momentum to the probe is negligible when compared to the probe's momentum [15]. It commands the evolution of the event-averaged probe quark distribution function $n(t, \mathbf{x}, \mathbf{p})$ as the probes traverse the medium.

$$\left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla - \nabla_p^\alpha X^{\alpha\beta}\nabla_p^\beta - \nabla_p^\alpha Y^\alpha\right) n(t, \mathbf{x}, \mathbf{p}) = 0 \tag{3.1}$$

where t is time, v is the probe's velocity, $\nabla^\alpha = (\partial_1, \partial_2, \partial_3)$ is the usual gradient, ∇_p^α is the gradient with respect to the probes' momentum, $\partial/(\partial p^\alpha)$, $\mathbf{x}$ and $\mathbf{p}$ are the probes' position and momentum, respectively, and α, β run from 1 through 3. The tensors $X^{\alpha\beta}$ and Y^α, from which $\hat{q}$ and $\mathrm{d}E/\mathrm{d}x$ will be calculated, are called the collision terms.

The collision terms are related to event-averaged[1] rates of change of the momentum of the probes over time:

$$\frac{\langle \Delta p^\alpha \Delta p^\beta \rangle}{\Delta t} = X^{\alpha\beta} + X^{\beta\alpha} \tag{3.2a}$$

$$\frac{\langle \Delta p^\alpha \rangle}{\Delta t} = -Y^\alpha \tag{3.2b}$$

As such, $\hat{q}$ and $\mathrm{d}E/\mathrm{d}x$ may be obtained from $X^{\alpha\beta}$ and Y^α:

$$\hat{q} = \frac{1}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right)\frac{\langle \Delta p^\alpha \Delta p^\beta \rangle}{\Delta t} = \frac{2}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right) X^{\alpha\beta} \tag{3.3a}$$

$$\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{v^\alpha}{v}\frac{\langle \Delta p^\alpha \rangle}{\Delta t} = -\frac{v^\alpha}{v} Y^\alpha \tag{3.3b}$$

Should these auxiliary calcs be moved to an appendix? To prove the first equality of eq. 3.3a,

$$\frac{1}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right)\frac{\langle \Delta p^\alpha \Delta p^\beta \rangle}{\Delta t} = \frac{1}{\Delta x}\left\langle (\Delta\mathbf{p})^2 - \left(\frac{\mathbf{v}\cdot\Delta\mathbf{p}}{v^2}\right)\right\rangle = \frac{\langle (\Delta\mathbf{p}_T)^2 \rangle}{\Delta x}$$

To prove the second equality of eq. 3.3a, eq. 3.2a is plugged in and the symmetry of $(\delta^{\alpha\beta} - v^\alpha v^\beta/v^2)$ under the switch of α and β is used.

[1] Requires further explanation if the event-averages haven't been explained so far.

To prove the first equality of eq. 3.3b, it is easiest to start from the definition of the stopping power:

$$\frac{\langle \Delta E\rangle}{\Delta x} = \frac{1}{v\Delta t}\left\langle \Delta(\sqrt{\mathbf{p}^2+m^2})\right\rangle = \frac{1}{v\Delta t}\left\langle \frac{\Delta(\mathbf{p}^2)}{2\sqrt{\mathbf{p}^2+m^2}}\right\rangle = \frac{1}{v\Delta t}\left\langle \frac{\mathbf{p}\cdot\Delta\mathbf{p}}{\sqrt{p^2+m^2}}\right\rangle = \frac{v^\alpha}{v}\frac{\Delta p^\alpha}{\Delta t}$$

$X^{\alpha\beta}$ is obtained using the event-averaged field correlators previously derived, while determining Y^α requires calculating the event-average of the correlator of one of the fields and each event's deviation of the quark-probe density from the event-averaged density, n, a quantity which is difficult to access.

In order to circumvent the challenges of calculating Y^α, it is assumed that the Fokker-Planck equation should, if the system were in equilibrium at a temperature T, admit the Boltzmann distribution function as a solution:

$$n^{\text{eq}}(\mathbf{p}) \propto \exp(E_{\mathbf{p}}/T), \qquad \text{with } E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + {m_Q}^2}$$

with m_Q the mass of the quark probe. This assumption allows obtaining Y^α from $X^{\alpha\beta}$ via the simple relation:

$$Y^\alpha = \frac{v^\beta}{T}X^{\alpha\beta} \tag{3.4}$$

For the purpose of obtaining Y^α from $X^{\alpha\beta}$, T is taken to be the temperature of a thermalized QGP with an energy density equal to that of the glasma[2]. That is, the temperature which would characterize the system achieved if the glasma were allowed to reach equilibrium without expanding. [Comments on validity?] - partly discussed in sec. 5 of Mrowczyncki_2018

From eqs. 3.3 and 3.4, the final expressions for determining $\hat{q}$ and $\mathrm{d}E/\,\mathrm{d}x$ from $X^{\alpha\beta}$ become:[3]

$$\hat{q} = \frac{2}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right)X^{\alpha\beta} \tag{3.5a}$$

$$\frac{\mathrm{d}E}{\mathrm{d}x} = -\frac{v}{T}\frac{v^\alpha v^\beta}{v^2}X^{\alpha\beta} \tag{3.5b}$$

As such, obtaining $\hat{q}$ and $\mathrm{d}E/\,\mathrm{d}x$ requires calculating the $\delta^{\alpha\beta}X^{\alpha\beta}$ and $v^\alpha v^\beta X^{\alpha\beta}$ projections of the $X^{\alpha\beta}$ tensor.

3.2 The $X^{\alpha\beta}$ tensor

The first collision term is obtained from the correlators of chromodynamic fields:[4]

$$\begin{aligned} X^{\alpha\beta}(t,\mathbf{x},\mathbf{v}) &= \frac{1}{N_c}\int_0^t \mathrm{d}t'\ \mathrm{Tr}_c\left[\left\langle F^\alpha(t,\mathbf{x})F^\beta(t',\mathbf{y})\right\rangle\right] \\ &= \frac{1}{2N_c}\int_0^t \mathrm{d}t'\left\langle F_a^\alpha(t,\mathbf{x})F_a^\beta(t',\mathbf{y})\right\rangle \end{aligned} \tag{3.6}$$

[2] The derivation of the glasma's energy density should end up before this section, so the reader will be familiar with the concept.
[3] The final form of dEdx will depend on wheteher the projection calculated will be on va*vb or va*vb/v**2
[4] The first line of this equation disagrees with [13] by a fator of $1/2$ but agrees with [15]. However, the second line is in agreement with [13]. This is presumably due to a typo in [13].

where N_c is the number of colors, Tr_c is a color trace in the fundamental representation, $\mathbf{y} = \mathbf{x} - \mathbf{v}(t - t')$, $F^\alpha \equiv F_a^\alpha t^a$[5] is the α component of the Lorentz color force, and a are color indices in the adjoint representation, which run from 1 through $N_c{}^2 - 1$. Thus, the $X^{\alpha\beta}$ component of the tensor is obtained by integrating, in time, over the full past trajectory of the probe, the correlator of the α and β components of the Lorentz color force; the first at the "present" (time t, position $\mathbf{x}$), and the second at a past point of the probe's trajectory (time t', position $\mathbf{y}$).

The Lorentz color force a quark is subject to when traveling with velocity $\mathbf{v}$ at time t and position $\mathbf{x}$ is:

$$\mathbf{F}_a(t,\mathbf{x}) = g\big(\mathbf{E}_a(t,\mathbf{x}) + \mathbf{v} \times \mathbf{B}_a(t,\mathbf{x})\big) \tag{3.7a}$$

$$F_a^\alpha(t,\mathbf{x}) = g\big(E_a^\alpha(t,\mathbf{x}) + \varepsilon^{\alpha\gamma\gamma'} v^\gamma B_a^{\gamma'}(t,\mathbf{x})\big) \tag{3.7b}$$

where g is the strong coupling constant and $\varepsilon^{\alpha\beta\gamma}$ is the Levi-Civita or antisymmetric tensor with $\varepsilon^{123} = 1$.

Plugging eq. 3.7b into eq. 3.6 allows writing the unfolded expression for $X^{\alpha\beta}$:

$$\begin{aligned} X^{\alpha\beta}(t,\mathbf{x},\mathbf{v}) = \frac{g^2}{2N_c}\int_0^t \mathrm{d}t' \Big[&\langle E_a^\alpha(t,\mathbf{x}) E_a^\beta(t-t',\mathbf{y})\rangle \\ &+ \varepsilon^{\beta\gamma\gamma'} v^\gamma \left\langle E_a^\alpha(t,\mathbf{x}) B_a^{\gamma'}(t-t',\mathbf{y})\right\rangle \\ &+ \varepsilon^{\alpha\gamma\gamma'} v^\gamma \left\langle B_a^{\gamma'}(t,\mathbf{x}) E_a^\beta(t-t',\mathbf{y})\right\rangle \\ &+ \varepsilon^{\alpha\gamma\gamma'}\varepsilon^{\beta\delta\delta'} v^\gamma v^\delta \left\langle B_a^{\gamma'}(t,\mathbf{x}) B_a^{\delta'}(t-t',\mathbf{y})\right\rangle \Big] \end{aligned} \tag{3.8}$$

[5] A proper appendix or introduction to SU(Nc) will be necessary

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Appendix A

Derivations

A.1 Covariant Current Conservation

$$[D_\nu, J^\nu] = [D_\nu, [D_\mu, F^{\mu\nu}]] = 0 \tag{A.1}$$

where the second equality may be shown by use of the Jacobi Identity:

$$\begin{aligned} &[D_\nu, [D_\mu, F^{\mu\nu}]] = -[D_\mu, [F^{\mu\nu}, D_\nu]] - [F^{\mu\nu}, [D_\nu, D_\mu]] \\ \Leftrightarrow\ &[D_\nu, [D_\mu, F^{\mu\nu}]] = -[D_\nu, [D_\mu, F^{\mu\nu}]] - ig[F^{\mu\nu}, F_{\mu\nu}] \\ \Leftrightarrow\ &2[D_\nu, [D_\mu, F^{\mu\nu}]] = 0 \end{aligned} \tag{A.2}$$

where, in the first step, the "dummy" indices of the first term on the right are switched ($\mu \leftrightarrow \nu$) and then the antisymmetry of the anticommutator and of the field strength tensor is used; for the second step, $[D_\mu, D_\nu]$ may be shown to equal $-igF_{\mu\nu}$:

$$\begin{aligned} [D_\mu, D_\nu] &= [\partial_\mu - igA_\mu, \partial_\nu - igA_\nu] = [\partial_\mu, \partial_\nu] - ig[\partial_\mu, A_\nu] - ig[A_\mu, \partial_\nu] - g^2[A^\mu, A^\nu] \\ &= 0 - ig\left(\partial_\mu A_\nu - A_\nu \partial_\mu + A_\mu \partial_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]\right) \\ &= -ig\left((\partial_\mu A_\nu) - (\partial_\nu A_\mu) - ig[A_\mu, A_\nu]\right) = -igF_{\mu\nu} \end{aligned} \tag{A.3}$$

and $[F^{\mu\nu}, F_{\mu\nu}] = 0$ because, first swapping the order in the commutator and then swapping the raised and lowered indices,

$$[F^{\mu\nu}, F_{\mu\nu}] = -[F_{\mu\nu}, F^{\mu\nu}] = -[F^{\mu\nu}, F_{\mu\nu}] \tag{A.4}$$

Appendix B

Coordinate Systems

B.1 Light-Cone Coordinates

May need to be reviewed so it has a similar structure and flow to the Bjorken coordinates appendix.

Throughout the derivations, it will be useful to work in light-cone coordinates. A four-vector $a^\mu = (a^0, a^1, a^2, a^3)$ may be expressed in light-cone coordinates as

$$a^\mu = (a^+, a^-, a^2, a^3) \equiv (a^+, a^-, \mathbf{a}_\perp), \qquad \mu = (+, -,\ 2,\ 3) \tag{B.1}$$

with

$$a^+ = \frac{a^0 + a^1}{\sqrt{2}} \qquad a^- = \frac{a^0 - a^1}{\sqrt{2}} \qquad \mathbf{a}_\perp = (a^2, a^3) \tag{B.2}$$

The dot product between two four-vectors a^μ and b^ν becomes

$$a \cdot b = a^+ b^- + a^- b^+ - \mathbf{a}_\perp \cdot \mathbf{b}_\perp = a^+ b^- + a^- b^+ - a^2 b^2 - a^3 b^3 \tag{B.3}$$

the metric being

$$g_{\mu\nu} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \tag{B.4}$$

The components of the corresponding covariant four-vector a_μ are

$$a_\mu = (a_+, a_-, a_2, a_3) = (a^-, a^+, -a^2, -a^3) \tag{B.5}$$

Considering the position four-vector,

$$x^\mu = (x^+, x^-, x^2, x^3) = \left(\frac{x^0 + x^1}{\sqrt{2}}, \frac{x^0 - x^1}{\sqrt{2}}, x^2, x^3 \right) \tag{B.6}$$

we may obtain the new base vectors. Focusing on the first two coordinates:

$$\vec{r} = x^0\vec{e}_0 + x^1\vec{e}_1, \quad \text{with} \quad \vec{e}_0 = (1,0,0,0), \quad \vec{e}_1 = (0,1,0,0) \tag{B.7}$$

$$\begin{aligned} \vec{e}_+ &= \frac{\partial\vec{r}}{\partial x^+} = \frac{1}{\sqrt{2}}\vec{e}_0 + \frac{1}{\sqrt{2}}\vec{e}_1 \\ \vec{e}_- &= \frac{\partial\vec{r}}{\partial x^-} = \frac{1}{\sqrt{2}}\vec{e}_0 - \frac{1}{\sqrt{2}}\vec{e}_1 \end{aligned} \tag{B.8}$$

Allowing us to rewrite $\vec{r}$:

$$\vec{r} = x^+\vec{e}_+ + x^-\vec{e}_- \tag{B.9}$$

Thus, the x^+ and x^- axes lie on the light cone (Fig. B.1).

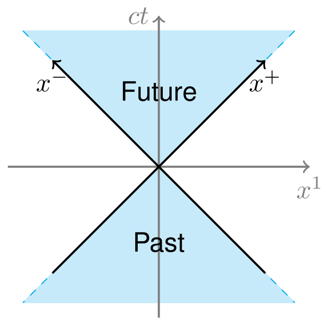

Figure B.1: Position four-vector axes in light-cone coordinates.

As for the four-gradient,

$$\partial_\mu = (\partial_+, \partial_-, \partial_2, \partial_3) = \left(\frac{\partial}{\partial x^+}, \frac{\partial}{\partial x^-}, \frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^3}\right) \tag{B.10}$$

$$\partial^\mu = (\partial^+, \partial^-, \partial^2, \partial^3) = (\partial_-, \partial_+, -\partial_2, -\partial_3) \tag{B.11}$$

Abusing language and the notation, one may similarly define the Dirac matrices "in light-cone coordinates".

$$\gamma^+ = \frac{\gamma^0 + \gamma^1}{\sqrt{2}} \qquad \gamma^- = \frac{\gamma^0 - \gamma^1}{\sqrt{2}} \tag{B.12}$$

They obey anticommutation relations of the same form as the usual Dirac matrices, with the new metric of Eq. B.4.

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}, \qquad \mu, \nu = (+, -, 2, 3) \tag{B.13}$$

B.2 Bjorken/Milne Coordinates

[1] In this system, two familiar quantities are used as the first two coordinates: proper time, τ of an object with $\mathbf{x}_\perp = 0$? which set off from the origin with constant velocity? – which is invariant under boosts

[1] what Margaret calls them on page 12

along the longitudinal direction, (x^1) –, and rapidity of some body in some particular circumstances too, η. The transverse coordinates are unchanged, as in light-cone coordinates.

$$\begin{aligned} \tau &= \sqrt{(x^0)^2 - (x^1)^2} = \sqrt{2x^+x^-} \\ \eta &= \frac{1}{2}\log\left(\frac{x^0+x^1}{x^0-x^1}\right) = \frac{1}{2}\log\left(\frac{x^+}{x^-}\right) \end{aligned} \tag{B.14}$$

As for the inverse transformation,

$$x^+ = \frac{\tau}{\sqrt{2}}e^{\eta} \qquad x^- = \frac{\tau}{\sqrt{2}}e^{-\eta} \tag{B.15}$$

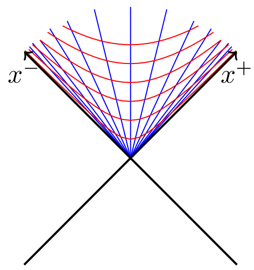

Figure B.2: Contour lines for proper time, τ, in red, and rapidity, η, in blue, in the first quadrant.

We may find the new basis vectors: versores n deviam ser "e"

$$\vec{r} = x^+\vec{e}_+ + x^-\vec{e}_-$$

$$\begin{aligned} \vec{e}_\tau &= \frac{\partial \vec{r}}{\partial \tau} = \frac{e^{\eta}}{\sqrt{2}}\vec{e}_+ + \frac{e^{-\eta}}{\sqrt{2}}\vec{e}_- = \frac{x^+}{\tau}\vec{e}_+ + \frac{x^-}{\tau}\vec{e}_- \\ \vec{e}_\eta &= \frac{\partial \vec{r}}{\partial \eta} = \frac{\tau}{\sqrt{2}}e^{\eta}\vec{e}_+ - \frac{\tau}{\sqrt{2}}e^{-\eta}\vec{e}_- = x^+\vec{e}_+ - x^-\vec{e}_- \end{aligned} \tag{B.16}$$

which are not, in general, of unit length.

By requiring

$$\vec{A} = A^+\vec{e}_+ + A^-\vec{e}_- \equiv A^\tau\vec{e}_\tau + A^\eta\vec{e}_\eta,$$

then

$$\begin{aligned} A^\tau &= \frac{x^-A^+ + x^+A^-}{\tau} \\ A^\eta &= \frac{x^-A^+ - x^+A^-}{\tau^2} \end{aligned} \tag{B.17}$$

Particularly,

$$\vec{r} = \tau\vec{e}_\tau \tag{B.18}$$